

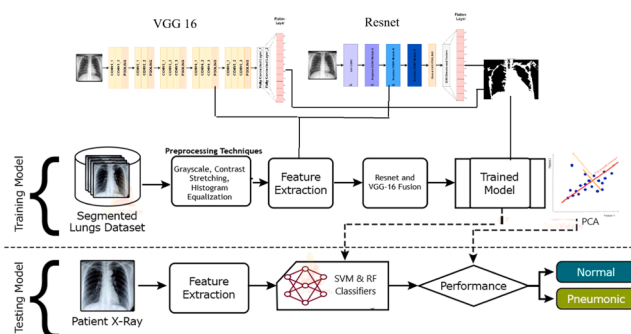


Hybrid deep learning and machine learning framework for automated pneumonia detection in chest X-ray images

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GRAPHICAL ABSTRACT



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ABSTRACT

Pneumonia is a dangerous respiratory illness that has to be precisely and promptly diagnosed in order to be treated effectively and prevent consequences. In order to distinguish between pneumonic and normal chest X-ray pictures, a hybrid deep learning technique is proposed in this study. For efficient and complementary feature extraction, the proposed system leverages the strengths of two popular convolutional neural networks, VGG16 and ResNet. Before training the model, the image is enhanced and the lungs are segmented by performing histogram equalisation, normalising contrast, and converting to grayscale. A richer feature representation of input photos is produced by fusing the features of VGG16 and ResNet. A model for identifying pneumonia is

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classified using the fused feature set. The system processes X-rays of new patients in order to extract features and categorise them using Random Forest (RF) and Support Vector Machine (SVM) classifiers. To increase accuracy and efficiency, feature dimensions are optimised using Principal Component Analysis (PCA). Key Contributions:

- 1. Dual-CNN feature fusion (VGG16 + ResNet) instead of single-model learning
- 2. PCA-based dimensionality optimization retaining 95% variance
- 3. Use of SVM and Random Forest for more interpretable diagnosis instead of CNN softmax

Specifications table

Subject area	Computer Science
More specific subject area	Medical Image Analysis, Deep Learning, Machine Learning
Name of your method	Hybrid CNN Feature Extraction with Dimensionality Reduction and ML Classification
Name and reference of the original method	VGG16, ResNet, PCA, SVM, Random Forest
Resource availability	Python, TensorFlow/Keras, Scikit-learn, OpenCV, Public X-ray Datasets

Background

The evolution of pneumonia identification in chest X-rays has transitioned from manual diagnosis to an automated AI-based system. Initial methods were limited to using hand-crafted features that would be combined with classic machine learning classifiers such as SVM [14], and Random Forest [15]. As deep learning started to gain popularity, researchers began using CNNs such as VGG16 [11], ResNets, and DenseNets that could learn the spatial features directly from the raw images [1]. With the existence of transfer learning, the pre-trained deep learning models could be fine-tuned on medical datasets to optimize the identification task. Eventually, attention mechanisms were introduced to encompass the Lung region and better model pneumonia disease [2]. Subsequently, Ensemble models would be proposed, further boosting robustness by drawing from the strength of the model architectures. It was very common that after training CNN models, high-dimensional features would need to be evaluated. Several researchers introduced dimensionality reduction techniques such as PCA to manage these high-dimensional MM features [3]. Other approaches fusing CNN-based feature extraction with machine learning classifiers would be termed hybrid approaches. Other researchers used clustering algorithms such as k-means to classify either disease severity levels or even categorize the distinguished outcomes [4]. Most recently, there is an increasing demand for explainable AI tools such as Grad-CAM and LIME that improve the latency behind the prediction and gain greater acceptance by the clinical community [5].

The process of image processing is essential in pneumonia detection, as it focuses on improving the quality of chest X-ray images and reducing difficulty in interpreting images. Most medical images are raw and can contain noise, artifacts, or low contrast, making certain pathological features challenging to detect [6]. Preprocessing processes such as turning the image to grayscale, binarization, and contrast enhancement make lung structures and abnormal areas more visible [7]. Threshold and morphological processing of images home in on areas of interest, such as nodules or kinks within the lungs, allowing for better analysis and interpretation of physiological information [8]. Segmentation allows the model to only focus on the lung fields while minimizing unwanted background information. These processes can reduce variant information presented from image to image, thereby allowing for improved consistency of training within deep learning models. When images are enhanced, more reliable means of feature extraction can occur with deep learning models [9]. Image processing can also lessen the computational efficiency of models by standardizing inputsImage processing is crucial to building reliable and efficient classification [16] algorithms, especially when combined with deep learning [17] convolutional neural networks (CNNs). This process allows diagnostic-based machine learning models to be more interpretable and clinically useful [10].

This method combines state-of-the-art deep learning (VGG16 and ResNet) [18,19] with considerable feature extraction, dimension reduction through Principal Component Analysis (PCA) [12,13], and a classifier that has machine learning capabilities (Support Vector Machine and Random Forest). This method resolves relevant problems that existing diagnostic systems face, which include noise in the imaging obtained, high dimensionality of features, and processing inefficiency.

Moreover, the component of the image preprocessing pipeline consisting of converting a color image to grayscale for the original nodule X-ray, thresholding a query candidate of a nodule, and ultimately also extracting a nodule, assures that the appropriate system can reliably discriminate "medically relevant" regions of the X-ray [20]. The proposed model can also capture novel and additional complex features. The exploration of features from three complementary CNN-architecture models (latent representation vectors) is compared to improve the overall model to provide more informative image features that are associated with pneumonia. The main goal of this approach is to create a scalable, interpretable, and highly reliable diagnosis tool that can help clinicians, accelerate decision-making, and improve patient outcomes in circumstances where access to experienced radiologists may be scarce.

Motivation

The bar graph in Fig. 2 depicts the number of deaths attributable to different diseases across the world, represented in millions. In

total, this highlights the burden imposed by pneumonia, as it is the most lethal disease caused that still remains completely preventable, highlighting the need for diagnostics and preventing risk within the general population. Lung cancer is below pneumonia in its death rate, at 0.92 million deaths. Other diseases are generating mortality at the 0.7 million mark.

Improving patient outcomes, particularly within environments with scarce assets, depends on the efficient and timely identification of pneumonia. Conventional diagnostic techniques rely frequently on the subjective and time-consuming manual interpretation of chest X-rays. This encourages the creation of automated tools that make use of cutting-edge deep learning and machine learning methodologies. By blending cutting-edge CNN architectures, like VGG16 and ResNet for fused feature extraction, and PCA for dimensionality reduction, this investigation seeks to address these issues. Machine learning classifiers such as SVM and Random Forest are used to further process the reduced feature set, guaranteeing precise pneumonia case recognition and categorisation into low, mild, and severe stages. Our method boosts image quality and concentrates on important regions of interest by using thorough preprocessing procedures. Our ultimate goal is to develop a scalable, dependable, and effective diagnostic system that aids physicians and optimises the rapidity and precision of patient diagnoses of pneumonia.

Previous works (Refer Table 1) primarily utilized single CNN models that did not leverage complementary fusion, did not include the necessary dimensionality reduction to address high-dimensional features, and few combined deep learning with interpretable classification using traditional machine learning classifiers. Our work fills these gaps through VGG16–ResNet feature fusion, optimizing for PCA, and hybridizing SVM and Random Forest for classification.

Method details

For pneumonia identification and severity classification that is feasible with the aid of a multi-stage pipeline (See Fig. 3) that converges upon deep extracted features combined with dimension reduction, image pre-processing, and a mixed set of machine/machine learning and clustering algorithms is suggested (See Fig. 1).

Dataset description and partitioning

The dataset employed in this research contains nearly 5000 chest X-ray images that were created for the automated detection and classification of pneumonia. Each image is subjected to a full preprocessing pipeline, including grayscale conversion, thresholding, and nodule extraction to highlight medically relevant locations while reducing noise. The dataset was partitioned into training, validation, and test sets to facilitate adequate model evaluation, while random translations, flips, zooming, and rotations were implemented as data augmentation during training to improve generalization. Sample chest X-ray images from the dataset showing both normal and pneumonia cases is given in Fig. 4. Table 2 gives the dataset description.

Image preprocessing

To reduce the input complexity from three channels to one, each raw chest X-ray image is first transformed from RGB to grayscale. To distinguish lung areas from the background, we then use Otsu's thresholding for automated binarization. To further refine these segmented areas, morphological operations such as erosion and dilation are used to remove small artifacts and enhance the lung

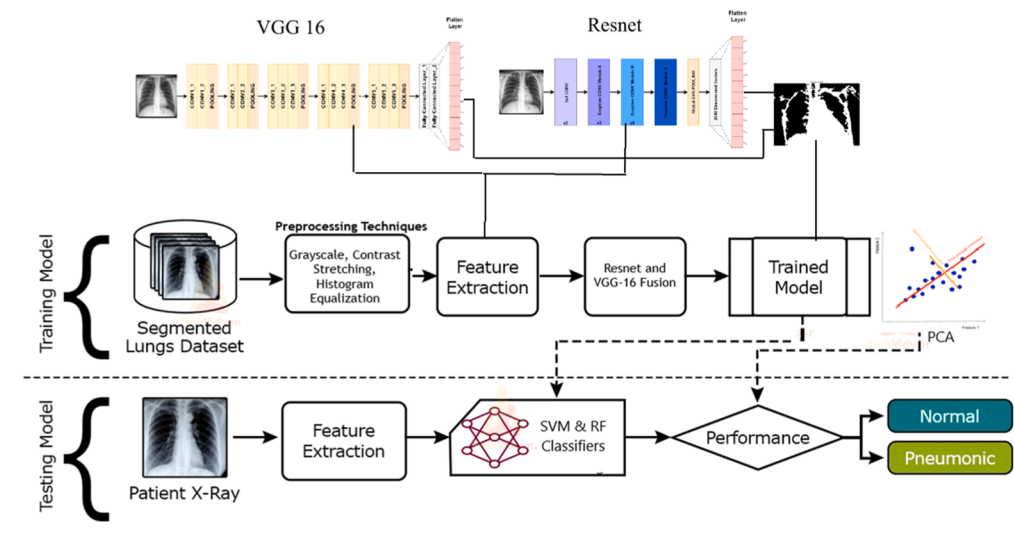


Fig. 1. Overview of the proposed hybrid framework combining dual-CNN feature fusion, dimensionality reduction, and interpretable machine learning classifiers for automated pneumonia detection from chest X-ray images.

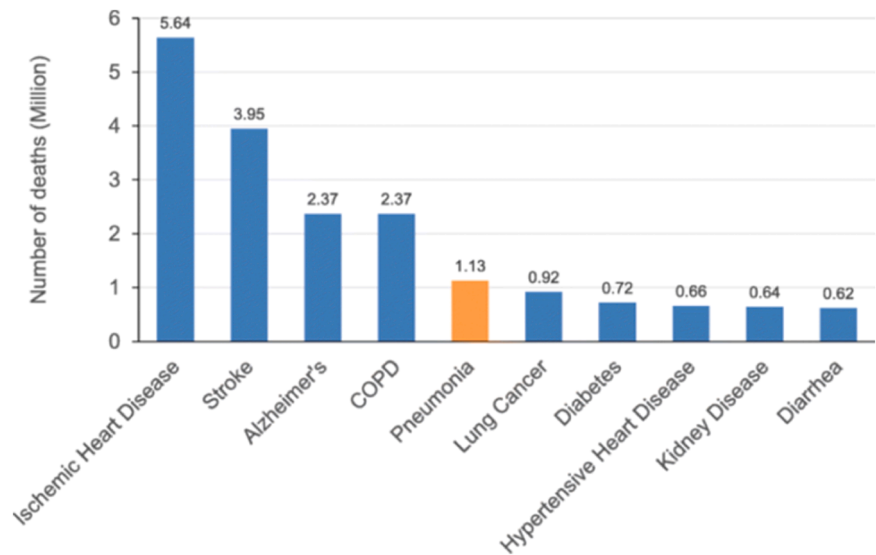


Fig. 2. Leading disease in the world.

Table 1
Comparison between the single-CNN approach and the proposed dual-CNN fusion framework highlighting key architectural, methodological, and performance differences.

Aspect	Sharma & Guleria (2023)	Proposed Hybrid Framework
Model Type	Single CNN (VGG16)	Dual CNN Fusion (VGG16 + ResNet)
Feature Extraction	Feature Extraction	Combined spatial + residual features via fusion
Dimensionality Handling	No feature reduction high-dimensional features retained	PCA applied to retain 95 % variance and reduce redundancy
Classifier	Fully connected CNN layers	Hybrid ML classifiers (SVM, Random Forest) for interpretable classification
Performance	Accuracy ≈ 95.2 %	Accuracy ≈ 98.8 % (SVM), 95.8 % (RF)

contours. Additionally, edge detection (using, for example, the Canny method) isolates fine structures, and region-growing algorithms extract nodular features that may indicate localized pneumonia infections. This multi-step preprocessing ensures that the network focuses on the lung fields, minimizing noise and irrelevant details.

Feature extraction

Preprocessed images are then fed into two pre-trained convolutional neural networks—VGG16 and ResNet. Both models, originally trained on ImageNet, are fine-tuned on our chest X-ray dataset. The feature extraction process is carried out by capturing outputs from the last convolutional blocks of both networks, after applying global average pooling, which encapsulates high-level, discriminative features. These features represent the complex textures, shapes, and lesion patterns present in pneumonia. The outputs of both VGG16 and ResNet are then combined to form fused feature extraction, which is a stronger and more detailed representation. Fusion is achieved by leveraging the complementary strengths of both architectures to obtain an enhanced representation of image features. Together, the feature representation forms a high-dimensional representation of each image that is used to extract detailed and high-level information that is critical for obtaining accurate pneumonia detection and classification. Segmented nodule region extracted from a chest X-ray image is illustrated in Fig. 5.

PCA based feature reduction

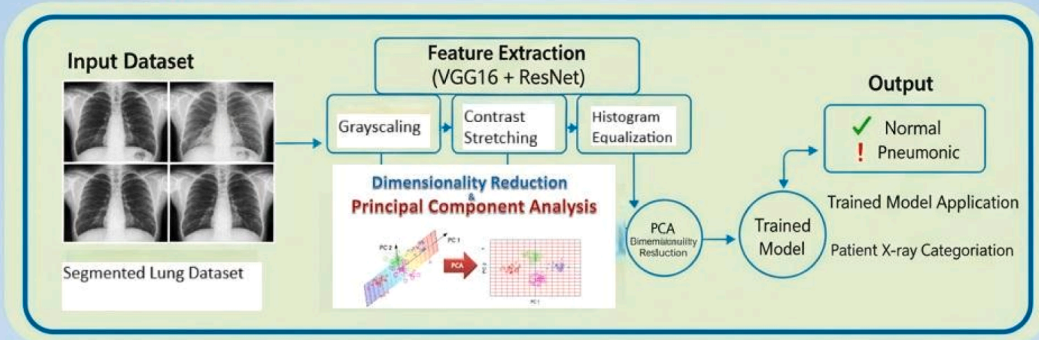
Since the concatenated feature vectors have a high dimension, Principal Component Analysis (PCA) is performed, which reduces the feature space to 100 principal components, which are chosen to preserve approximately 95 % of the variance. The important benefit of this step is that the subsequent computation becomes easier, and the likelihood of overfitting is reduced since redundant information is eliminated. PCA-based feature reduction to improve efficiency while retaining diagnostic information.

$$Z = W^T (X - \mu)$$

Where:

Pneumonia Detection System Architecture

1. Training Model Phase



2. Testing Model Phase

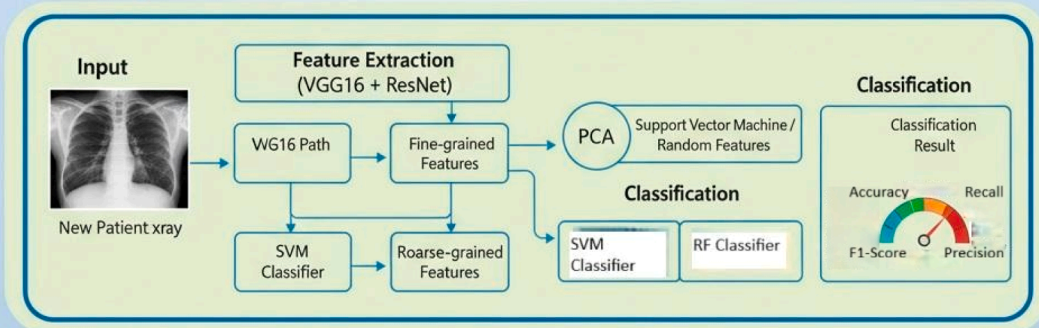


Fig. 3. Step-by-step workflow of the proposed pneumonia detection system.

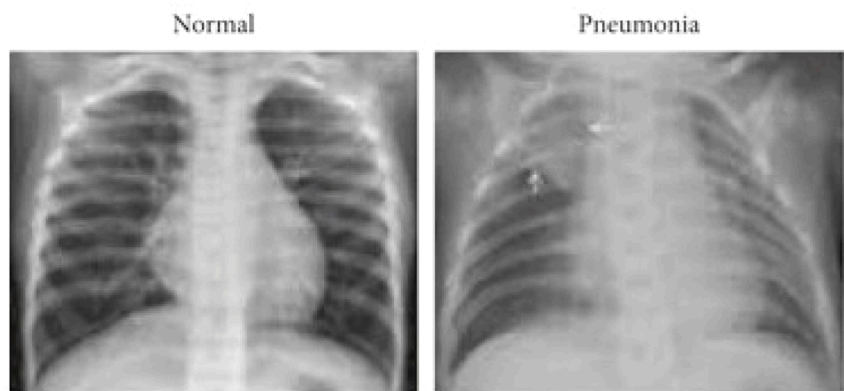


Fig. 4. Normal vs pneumonia chest.

Table 2
Dataset description table.

Attribute	Description
Dataset Name	Chest X-ray Pneumonia Dataset
Source	https://www.kaggle.com/datasets/andrewmvd/pediatric-pneumonia-chest-xray
Total Number of Images	5856
Image Modality	Chest X-ray (CXR)
Image Categories	- Normal - Pneumonia
Image Format	JPEG or PNG (standard X-ray image formats)



Fig. 5. Nodule image.

X is the feature matrix, μ is the mean vector.

W contains the top k eigenvectors of the covariance matrix $C = \frac{1}{n}(X - \mu)^T(X - \mu)$

Z is the reduced-dimensional representation used for classification.

Dual-network feature fusion (VGG16 + ResNet)

Feature fusion is used for enriched spatial and residual learning representation.

$$F_{fusion} = \alpha F_{VGG16} + (1 - \alpha)F_{ResNet}$$

Where:

$F_{\{ResNet\}}$ and $F_{\{VGG16\}}$ are the global average pooled feature vectors from their respective final convolutional blocks.

$\alpha \in [0, 1]$ is a weighting factor determined experimentally.

Classification

The reduced feature vectors are used to train two classifiers: Support Vector Machine (SVM) with an RBF kernel and a Random Forest (RF) model. Both the hyperparameters of both models are optimized by cross-validation to optimize a balanced performance. Clinical guidelines require several clusters, and clinical severity scores are used in cluster centroids calibration against threshold values.

A hybrid framework

By combining CNN feature extraction with classical machine learning classifiers the hybrid framework integrates learned representations F_{CNN} with efficient decision boundaries of machine learning models.

$$\text{Final Prediction : } \hat{y} = \text{inf}_{ML(PCA(F_{CNN}))}$$

Where:

$F_{\{CNN\}} = [F_{\{VGG16\}} || F_{\{ResNet\}}]$ is the concatenated deep feature vector.

$f_{\{ML\}}$ denotes SVM or Random Forest classifier.
PCA reduces dimensionality while preserving variance.

Training and evaluation

Subsets of datasets are used for testing, validation, and training. Data augmentation methods are applied for training to increase resilience (random translations, flips, zooming, and rotations). Example metrics used for measuring the performance of the model are accuracy, precision, recall, F1-score, and AUC-ROC. This structured approach makes it possible to diagnose and stage the severity of pneumonia in a precise, efficient, and interpretable way.

Pseudocode

Input: Chest X-ray images $I = \{I_1, I_2, \dots, I_n\}$

Output: Predicted label $y \in \{Normal, Pneumonia\}$

Step 1: Image Preprocessing

- Perform histogram equalization and contrast stretching to enhance visibility.
- Obtain pre-processed image I_p .

Step 2: Dual-Network Feature Extraction

- Feed I_p into VGG16 and ResNet models pre-trained on ImageNet.
- Extract deep features:

$$F_{VGG16} = f_{VGG16}(I_p)$$

$$F_{ResNet} = f_{ResNet}(I_p)$$

- Fuse spatial and residual features:

$$F_{fusion} = \alpha F_{VGG16} + (1 - \alpha) F_{ResNet}$$

Step 3: Dimensionality Reduction using PCA

- Compute covariance matrix:

$$C = (1/n) \times (X - \mu)^T (X - \mu)$$

- Compute eigenvectors W of C .
- Project fused features onto principal components:

$$Z = W^T (F_{fusion} - \mu)$$

- Retain top k components preserving $\approx 95\%$ variance.

Step 4: Classification

- **Support Vector Machine (SVM):**

Minimize

$$1/2w^2 + C \sum \xi_i$$

subject to $y_i (w^T \phi(x_i) + b) \geq 1 - \xi_i, \xi_i \geq 0$

Kernel: $K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$

K is the kernel function used in SVM to compute similarity between samples for classification.

- **Random Forest (RF)**

$$\hat{y} = \text{mode}\{h_t(x), t=1, 2, 3, \dots, T\}$$

Step 5: Prediction and Evaluation

1. For each test image I_{test} , extract and reduce features $\rightarrow Z_{\text{test}}$.
2. Predict class label:

$$\hat{y} = f_{\text{ML}}(Z_{\text{test}})$$

3. Evaluate using Accuracy, Precision, Recall, F1-score, and AUC-ROC.

Method validation

Preprocessing validation

Grayscale picture conversion as seen in Fig. 6 is an important preprocessing step for certain application in medicine; its reduction of noise and processing cost can be achieved by combining large amounts of color knowledge into one intensity channel, that of brightness; thus simply looking only at brightness changes, it captures relevant information and makes the further analysis more precise, for example by detecting minute irregularities in chest X-rays to rule out pneumonia in clinical examination.

Histogram equalization is an enhancement technique used for chest X-ray image processing to increase the contrast of structures. It redistributes the values of pixels of their intensity so that the full intensity range is available for imaging, and dark areas are made lighter and light areas darker. This increases contrast between the lung and bone tissues and helps in the diagnosis and analysis of abnormalities. Histogram equalization used to enhance contrast and highlight lung structures is depicted in Fig. 7.

Contrast stretching is an image enhancement technique that increases the range of intensity values in chest X-ray images by mapping the input pixel values to a broader range. This increases the contrast between dark and light areas in the images. Crossing the median contrast between the dark and white areas helps to highlight important anatomical structures and abnormalities in the images and makes them easier to interpret later in the medical career. Contrast stretching applied to improve intensity distribution in chest X-ray images is shown in Fig. 8.

Preprocessing image segmentation is the process of pre-processing an image to organize it into sections relevant to the processing application. It can be applied to medical imaging to make precise diagnoses of problems with structures or anomalies, and remote sensing to differentiate different types of land cover. Segmented lung mask showing isolated lung regions after preprocessing is shown

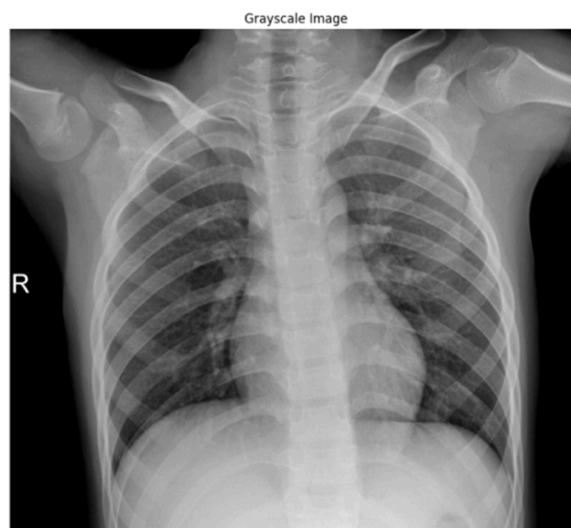


Fig. 6. Grayscale image.



Fig. 7. Histogram equalization.

in Fig. 9. Segmentation reduces analysis time, noise, and overall processing speed for subsequent processing tasks by highlighting relevant structures.

Nodule extraction is a procedure in medical imaging that focuses attention on small, round structures called nodules in a lung CT scan. This technique makes further evaluation more accessible by separating potential nodules and ensuring important features are accurately identified. Example of nodule extraction highlighting potential abnormal lung regions is shown in Fig. 10. Nodule extraction increases diagnosis accuracy in finding abnormalities, and automation systems can help detect things like pneumonia.

Feature extraction validation

In this study, the feature extraction process is the most important part of the study, which comprises the use of deep learning models to identify and classify pneumonia from chest X-ray images. For feature extraction, two pretrained convolutional neural networks, VGG16 and ResNet, are used with the chest X-ray data using models that were originally trained using ImageNet data. The authors extracted features using the output from the last convolutional blocks of the networks after applying global average pooling. The authors indicated that this will retain the most discriminative, high-level features that represent complicated textures and lesion characteristics vital to pneumonia.

To improve representation power, features from VGG16 and ResNet are fused by concatenation to combine the spatial sensitivity of VGG16 with the residual learning capability of ResNet. Fused features are a high-dimensional representation where diagnostic information of the images will be adequately represented.

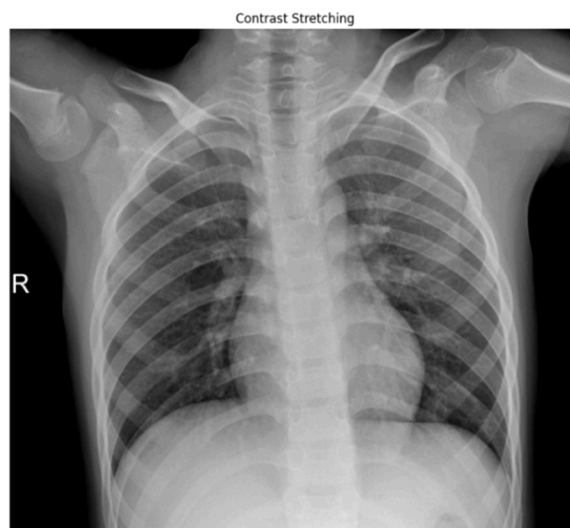


Fig. 8. Contrast stretching.

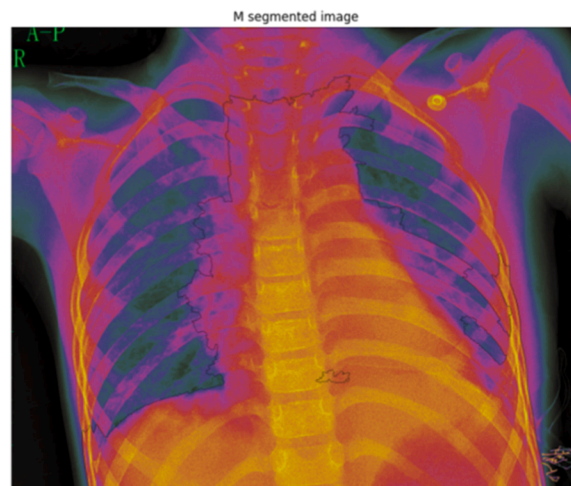


Fig. 9. M-segment image.

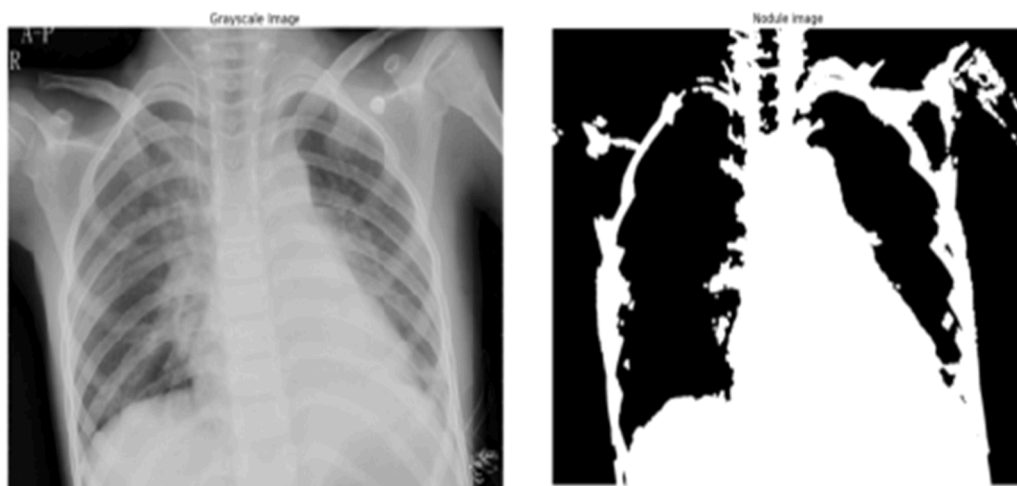


Fig. 10. Nodule image.

Dimensionality reduction validation

In the perspective framework, dimension reduction is an integral task that will enhance the performance and efficiency of the pneumonia detection system. After obtaining high-dimensional feature vectors from chest X-ray images from the fused outputs of two self-supervised pre-trained convolutional neural networks (VGG16 and ResNet), the authors apply Principal Component Analysis (PCA) to condense the feature space. This is an important operation in terms of mitigating the inherent difficulties in working with high-dimensional data (specifically, the increased computational cost and overfitting).

PCA is utilized to keep 100 principal components, with 100 carefully selected principal components retaining approximately 95 % of the total variance of the data. All the principal components indicate the informative attributes of the image data and the most diagnostically relevant components, while removing the redundant and less important components. The reduced feature space allows the classifiers – Support Vector Machine (SVM) and Random Forest (RF)—to be trained more robustly and efficiently.

Classification and hyperparameter tuning

The proposed approach to automatic pneumonia detection incorporates two supervised machine learning classifiers, Support Vector Machine (SVM) and Random Forest (RF), which classify the deep features that have been extracted from VGG16 and ResNet and reduced to 100 principal components through PCA. The classifiers use the refined deep features as inputs. In the case of SVM, a Radial Basis Function (RBF) kernel was employed, which allows SVM to work effectively on non-linear distributions of data and works well on the dataset, producing high performance. The Random Forest classifier works via ensemble learning with multiple decision trees, and

performs quite well on the dataset also Both classification algorithms were hyperparameter tuned using cross-validation so as to optimize the algorithm performance and to avoid overfitting through values of the actual parameter.

Performance evaluation metrics

Performance assessment of the hybrid deep and machine learning framework for pneumonia detection implemented here is performed using multiple well-recognized metrics. Accuracy, precision, recall, F1-score, and AUC-ROC (Area Under the Receiver Operating Characteristic Curve) can be calculated using a confusion matrix that shows the true positives, false positives, true negatives, and false negatives, which helps us see the strengths and weaknesses of the model. In summary, the SVM classifier had an accuracy of 98.8 %, with an associated precision of 0.99, a recall of 0.98, and an associated F1-score of 0.98, indicating a good predictive ability with an exceedingly low expected false positive rate and false negative rate. The RF classifier had an accuracy of 95.8 %, an associated precision of 0.96, a recall of 0.94, and an associated F1-score of 0.95. AUC-ROC of the classifiers reached values of 0.92, which indicates a strong discriminative ability between pneumonia and non-pneumonia classes. Confusion metrics refer to metrics that quantify the performance of a classifier and are usually stored in a confusion matrix. Confusion metrics show the misfit in the prediction model by counting actual positives, exaggerated positives, real negatives, and fake negatives. The metrics on the confusion matrix, including accuracy, precision, recall, and F1-score, can be used to identify the benefits and shortcomings of the classifier, to inform improvements, and to achieve more reliable models.

Figs. 11 and 12 gives the confusion matrix of the SVM and Random Forest classifiers respectively, illustrating prediction accuracy. As claimed accuracy of approximately 99 %, the SVM classifier has shown almost perfect accuracy in detecting pneumonia from normal X-ray images: 12 pneumonia cases were mistakenly classified as normal, 4 normal cases were misclassified as pneumonia according to the confusion matrix. Clinical effectiveness of the SVM method is reflected by the high accuracy (precision/recall) and F1-scores of our method, which guarantee reliable results, as well as patient safety margins for better results.

Here, we evaluate the performance of the above Random Forest classifier to detect pneumonia in chest X-ray images with about 95.8 % accuracy. Confusion matrix: There are only a few misclassifications, with 27 normal images classified as pneumonia and 31 pneumonia images classified as normal. The F1-scores of both classes are high, which demonstrates the effectiveness of the model for reliable diagnosis and better patient care in clinical settings.

The ROC [for True Positive Rate and False Positive Rate] plots the performance of the classification for the two data sets, and, with a higher curve representing better classification performance, the AUC-ROC [for Analysis Unit for Randomized Covariance] = 0.92. Fig. 13 depicts ROC curve classifier discrimination ability with AUC-ROC value. The example output of the trained model classifying a test X-ray image is given in Fig. 14.

Fig. 15. Illustrates a radar chart, a multifaceted visualisation method that successfully shows a model's performance across a number of assessment measures is the radar chart. Five standard metrics—Accuracy, Precision, Recall, F1-Score, and AUC-ROC—were employed in this context to assess the performance of Random Forest and Support Vector Machine (SVM) classifiers. The radar chart's displayed lines create a closed shape, allowing for rapid visual interpretation. Each axis indicates a single measure. A model with

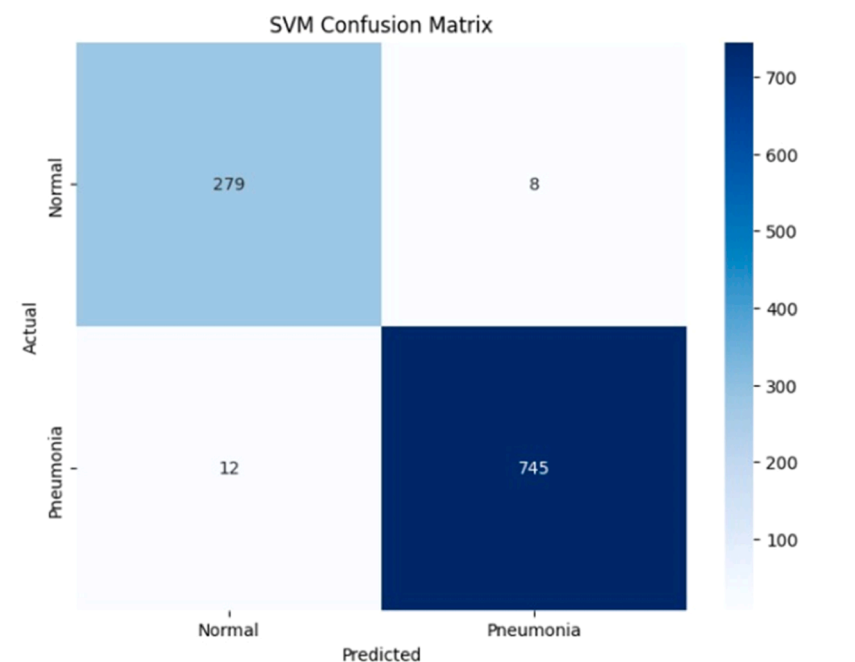


Fig. 11. SVM confusion metrics.

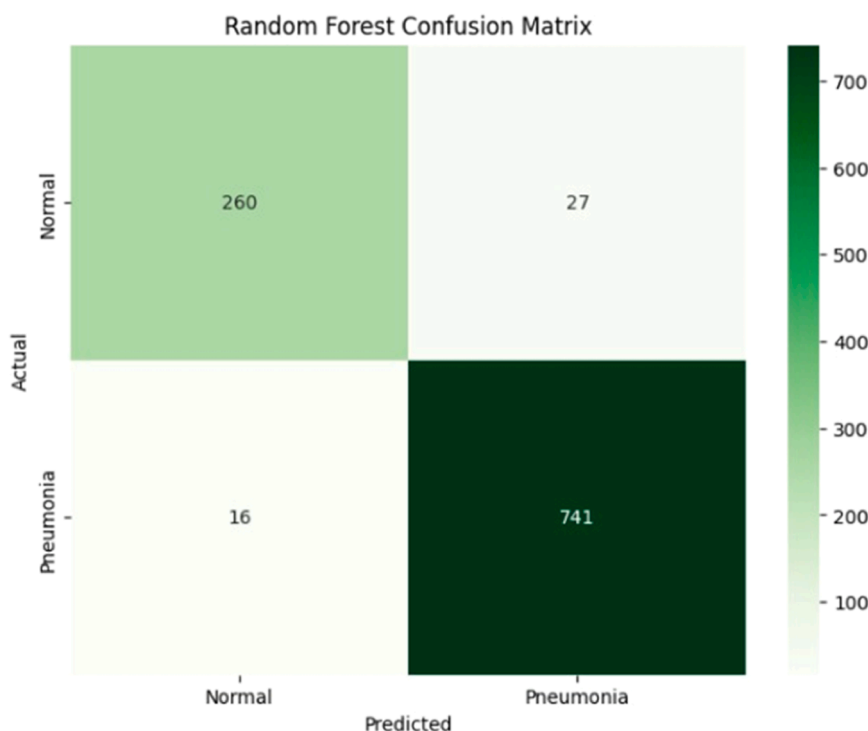


Fig. 12. RF confusion metrics.

consistently good results across all measures is shown by an outward-stretching, nearly complete shape. Finding relative strengths and weaknesses is made simpler by superimposing the shapes of several models on a single graphic. For example, a model's shape will include a greater region if it continuously performs better than another in several dimensions.

By determining how each metric score deviates from the optimal value of 100 %, the performance gap heatmap in Fig. 16 offers a targeted perspective on model flaws. It is an effective diagnostic tool since it shows the areas in which each model fails rather than highlighting its flawless performance. The error (or "gap") in a certain measure for a particular model is represented by each cell in the heatmap, with colour intensity indicating the size of the disparity. A deeper hue, for instance, would suggest that the model has more memory issues than precision issues. The heatmap provides a comparison viewpoint by arranging two models side by side, which aids in better decision-making, especially when enhancing or optimising categorisation systems.

Table 3 provides a comparative analysis of several deep learning-based methods for image classification tasks that have been put out in recent research, emphasising the techniques used and the accompanying accuracy levels. With an accuracy of 98.8 %, the suggested model in this study (2025) fared better than any other approach currently in use.

Clinical interpretation

The performance metrics of the proposed framework are not only statistically strong but also meaningful when viewed from a clinical decision-making perspective. In the context of pneumonia, delayed or missed diagnosis can rapidly lead to complications such as respiratory distress, septic shock, and increased mortality. Therefore, the usefulness of an AI-assisted diagnostic tool must be evaluated based on how well it reduces clinical risk, rather than on numerical accuracy alone.

The Support Vector Machine (SVM) classifier recorded a recall of 0.98, which indicates that almost every pneumonia case in the test set was correctly detected. This is clinically crucial because a false negative (undiagnosed pneumonia) carries significantly higher risk than a false positive. When an infected patient is incorrectly classified as "normal," antibiotic treatment, chest physiotherapy, or hospitalization may be postponed, worsening the patient's prognosis. Thus, the model's bias toward higher recall directly supports safer clinical decision pathways.

Likewise, the model achieved a precision value of 0.99, meaning that very few healthy patients were incorrectly labeled as having pneumonia. While false positives are less dangerous from a medical standpoint, they still impact healthcare by triggering unnecessary imaging reviews, medication prescriptions, and patient anxiety. The high precision therefore reduces clinical workload and resource consumption, which is especially beneficial in overburdened healthcare environments.

The confusion matrix analysis (Figs. 11 and 12) confirms that both misclassification types are minimized, with only a small number of errors across thousands of images. This level of reliability suggests that the system can act as a supportive screening tool for radiologists, particularly in emergency departments, remote clinics, or screening camps where expert radiologists may not always be present.

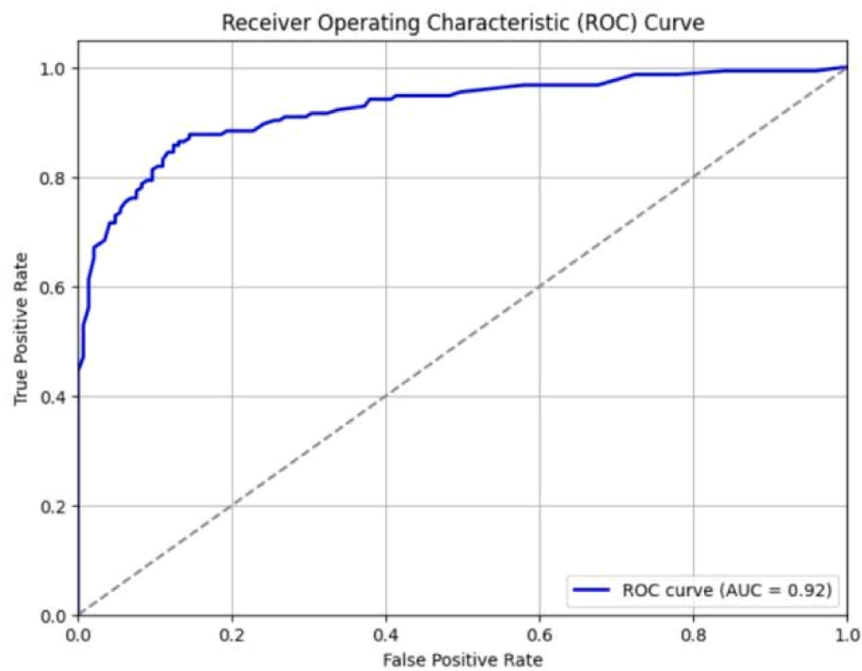


Fig. 13. AUC-ROC (area under the receiver operating characteristic curve).



Fig. 14. Result of the input image.

Another practical benefit is the use of PCA, which reduces the feature dimensions while retaining 95 % of meaningful variation. This not only improves computational speed but also shortens model inference time—an essential requirement when decisions must be made rapidly, such as during outbreaks or in point-of-care settings.

Finally, the inclusion of interpretable machine learning models (SVM and Random Forest), rather than a black-box CNN softmax output, allows decision boundaries to be examined and justified. This aligns with current expectations in clinical AI deployment, where explainability and traceability are increasingly required by medical regulators and hospital ethics committees. Table 4 describes the clinical relevance mapped to model insight.

Limitations

Dependence on image quality and homogeneity

The framework presupposes a minimum quality standard of input chest X-ray images. In cases of low-quality images (e.g., blurred, underexposed, or overexposed scans), there is the potential of failure in the preprocessing steps of thresholding and nodule extraction, which will not reliably segment lung regions.

Differences in images derived from different X-ray machines or clinical settings with differing (e.g., scanner-dependent, or clinical

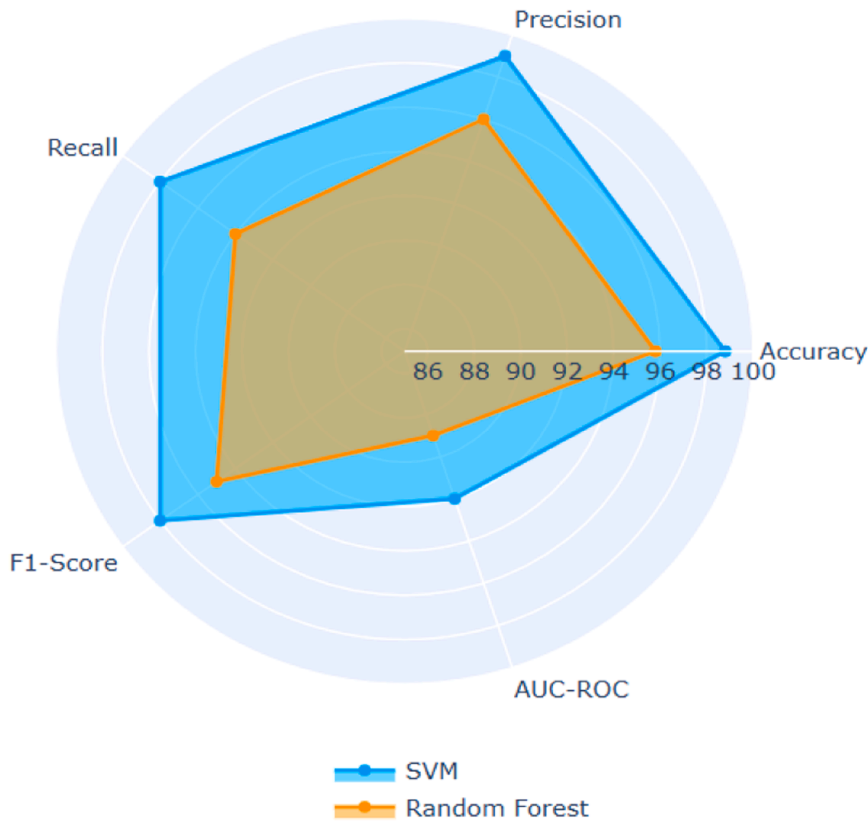


Fig. 15. Radar chart for model performance.

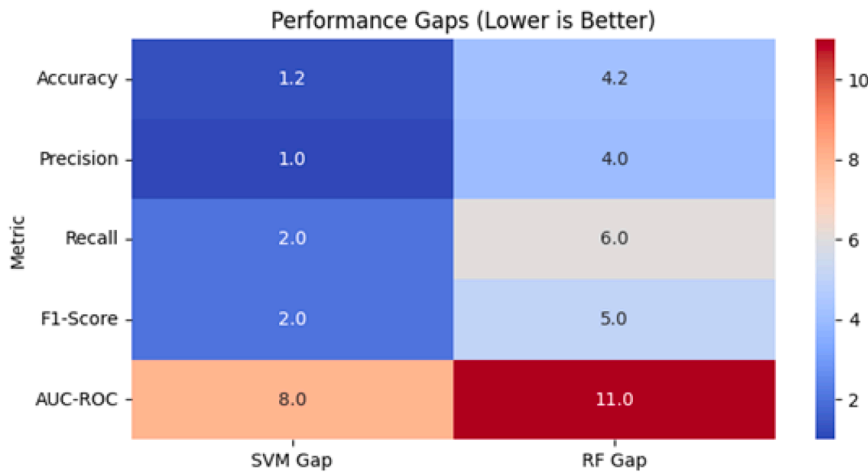


Fig. 16. Performance gaps.

Table 3
Comparative analysis.

Author(s)	Year	Methodology Used	Accuracy
Qiuyu An et al.	2024	Attention-Ensemble Deep CNN	95 %
Roy Ayush et al.	2024	FA-Net	97 %
Mabrouk et al.	2023	Ensemble of CNNs + Vision Transformer	93.91 %
Proposed Model	2025	Image preprocessing + CNN with multiple Conv layers	98.8 %

Table 4
Clinical significance summary.

Model Insight	Clinical Relevance
High Recall (0.98)	Prevents missed pneumonia cases (high-risk outcome)
High Precision (0.99)	Reduces unnecessary follow-ups & antibiotic misuse
PCA-based Feature Reduction	Faster screening, deployable in emergency triage
Hybrid ML Classifier	Transparent decisioning enables clinical auditability
Low False Negative Rate	Aligns with WHO and CDC pneumonia screening priorities

post-processing) resolutions and modalities may cause variation that could compromise model quality.

Sensitivity to contrast enhancement parameters

While there are some processes, such as Histogram Equalization and Contrast Stretching, which work effectively to increase visual contrast, there is the potential to over-amplify noise or falsely-defined structures in any of the individual scans, particularly in children or elderly patients who have differing anatomical considerations.

In circumstances where lung opacities are faint or diffused, such models would amplify the proximal structures, could be misleading, and distort relevant regions in a manner that would jeopardize feature extraction.

Limited generalizability across varied clinical datasets

The model was trained on a dataset of 5000 chest X-rays. The dataset size is decent; however, it may not adequately capture the entirety of the variability of the populations of patients (e.g., age, sex, comorbidities), appreciate the various pathologies (e.g., viral pneumonia or bacterial pneumonia), or administer treatment uniquely based on regional clinical practices.

To apply to datasets from populations with different characteristics or use typical pneumonia presentations may render the qualities of the model less accurate, leading to false-positive or false-negative results.

Lack of explainability and clinical interpretability

Despite fused features from VGG16 and ResNet being highly discriminative, they are still “black box” representations. The model does not currently have any explainable AI tools (e.g., Grad-CAM or SHAP) to support clinical trust and usability. The issue of explainability and validation is especially critical in medical applications, where visual explanation of a diagnosis is essential for physicians’ corroboration and decision-making.

Performance in overlapping and multi-label cases

The present classification system is binary (pneumonia vs. non-pneumonia), and the severity stage is determined via k-means clustering. The machine can not be certain of multi-label scenarios (e.g., tuberculosis, fibrosis, co-occurrence, or overlaps) of thoracic pathologies. Some false classifications may likely occur when there are multiple thoracic pathologies because relevant feature patterns will overlap.

Computational requirements

The dimensionality of data will be reduced first through PCA; however, both the initial feature extraction through a CNN, as well as the training of the model, will use significant computational capacity and GPU acceleration.

This may hinder deployment in low-resource or point-of-care settings, particularly in rural health contexts.

Conclusion

This study presents a hybrid deep learning and machine learning framework that combines dual-CNN feature fusion, dimensionality reduction, and interpretable classification for automated pneumonia detection from chest X-ray images. Unlike single-network approaches, the proposed method integrates VGG16 and ResNet feature representations, followed by PCA to reduce redundancy while preserving 95 % of the variance. The reduced feature vectors are then classified using SVM and Random Forest, achieving accuracies of 98.8 % and 95.8 %, respectively.

The major contribution of this work is the shift from end-to-end CNN classification toward a more explainable and computationally efficient pipeline that supports clinical decision-making. The high recall score of the SVM model minimizes the risk of missed pneumonia cases, while the use of interpretable classifiers provides a transparent diagnostic pathway suitable for real-world medical environments. Future work will focus on extending the framework to multi-class disease classification, incorporating explainable AI tools such as Grad-CAM or SHAP, and validating the model on multi-center clinical datasets to ensure generalizability across diverse imaging conditions.

Ethics statements

a) informed consent was obtained from participants or that participant data has been fully anonymized, and b) the platform(s)' data redistribution policies were complied with.

CRediT author statement

Akshay S: Conceptualization, Methodology, Draft preparation, Review. **Sneha Kashyap:** Supervision, Writing- Reviewing and Editing. **Danikk Patel H N:** Validity tests, Data curation, Writing- Original draft preparation. **Pavan Kumar Y R:** Visualization, Investigation, Writing- Reviewing and Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data is publicly available at <https://www.kaggle.com/datasets/andrewmvd/pediatric-pneumonia-chest-xray>.

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